

Review of Test 2.

* Chapter 8. The Sherman-Morrison Formula.

A, B two $n \times n$ matrices related by

$$A = B - UV^t$$

Where $u \neq 0, v \neq 0$ are column n -vectors. Assume B is nonsingular.

1. A is also nonsingular if and only if

$$v^t(B^{-1}u) \neq 1.$$

2. If A is nonsingular, then A^{-1} is given by

$$A^{-1} = \left[I + \left(\frac{1}{1 - v^t(B^{-1}u)} \right) (B^{-1}u)v^t \right] B^{-1}$$

Solution Process of $AX=b$ by the Shorr. Formula

Matrix: $A = B - UV^t$

Equation: $AX=b$

Algorithm:

step 1. Solve $BZ=u$

step 2. Compute $\gamma = 1 - v^t Z$

If $\gamma = 0$, then A is singular and stop

If $\gamma \neq 0$, then A is nonsingular and continue.

step 3. Solve $BY=b$

step 4. Compute $\beta = (v^t y) / \gamma$

step 5. Compute $x = y + \beta Z$

* Chapter 9. Vector and Matrix Norms.

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

vector Norm:

$$\|X\|_1 = \sum_{k=1}^n |x_k|$$

$$\|X\|_2 = \left(\sum_{k=1}^n x_k^2 \right)^{\frac{1}{2}}$$

$$\|X\|_{\infty} = \max_{1 \leq k \leq n} |x_k|.$$

Properties of Norm $\|\cdot\|$.

(1) Positivity: $\|X\| \geq 0$ and $\|X\| = 0$ iff $X = 0$

(2) Homogeneity: $\|\alpha X\| = |\alpha| \|X\|$

(3) Triangle Inequality: $\|X+Y\| \leq \|X\| + \|Y\|$

Cauchy - Schwartz Inequality:

$$|X^T Y| \leq \|X\|_2 \|Y\|_2$$

Relations among $\|\cdot\|_1$, $\|\cdot\|_2$ and $\|\cdot\|_{\infty}$ for vectors.

$$\|X\|_2 \leq \|X\|_1 \leq \sqrt{n} \|X\|_2$$

$$\|X\|_{\infty} \leq \|X\|_2 \leq \sqrt{n} \|X\|_{\infty}$$

$$\|X\|_{\infty} \leq \|X\|_1 \leq n \|X\|_{\infty}$$

Matrix Norm:

A an $n \times n$ matrix, then $\|A\| = \max_{X \neq 0} \frac{\|AX\|}{\|X\|}$.

Properties:

1. $\|A\| \geq 0$ and $\|A\| = 0$ iff $A = 0$

2. $\|\alpha A\| = |\alpha| \|A\|$

5. $\|AX\| \leq \|A\| \|X\|$

3. $\|A+B\| \leq \|A\| + \|B\|$

6. $\|AB\| \leq \|A\| \|B\|$

4. $\|I\| = 1$

Explicit Formula:

$\|A\|_1$ = maximum of the absolute column sums.

$\|A\|_\infty$ = maximum of the absolute row sums.

Relations among $\| \cdot \|_1$, $\| \cdot \|_2$, $\| \cdot \|_\infty$ for matrices.

$$\frac{1}{\sqrt{n}} \|A\|_2 \leq \|A\|_1 \leq \sqrt{n} \|A\|_2.$$

$$\frac{1}{\sqrt{n}} \|A\|_\infty \leq \|A\|_2 \leq \sqrt{n} \|A\|_\infty$$

$$\frac{1}{n} \|A\|_\infty \leq \|A\|_1 \leq n \|A\|_\infty$$

* Chapter 10. Condition Number,

$$K(A) = \|A\| \cdot \|A^{-1}\|.$$

Properties:

1. $K(I) = 1$

2. $K(A) \geq 1$

3. $K(AB) \leq K(A) K(B)$

4. $K(\alpha A) = K(A)$ for $\alpha \neq 0$

5. $K(A) = K(A^{-1})$.

Relative error of $AX=b$

$$\boxed{\frac{\|x - x'\|}{\|x\|}} \leq K(A) \frac{\|b - b'\|}{\|b\|}$$

where x' is an approximate solution and $b' = Ax'$
significant digits (definition).

* Chapter 11. Iterative method.

$$Sx^{\text{new}} = b + Tx^{\text{old}} \quad (Ax=b, A=S-T)$$

for $k=0, 1, 2, \dots$

$$Sx_{k+1} = b + Tx_k.$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

The Jacobi method.

$$S = \begin{bmatrix} a_{11} & & & \\ & a_{22} & & \\ & & \ddots & \\ & & & a_{nn} \end{bmatrix} \quad T = - \begin{bmatrix} 0 & a_{12} & \dots & a_{1n} \\ a_{21} & 0 & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & 0 \end{bmatrix}$$

The Gauss-Seidel Method.

$$S = \begin{bmatrix} a_{11} & & & \\ a_{21} & a_{22} & & \\ \vdots & \vdots & \ddots & \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \quad T = - \begin{bmatrix} 0 & a_{12} & \dots & a_{1n} \\ & 0 & \dots & a_{2n} \\ & & \ddots & \vdots \\ & & & 0 \end{bmatrix}$$

Example: $\begin{bmatrix} 4 & 1 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

Jacobi Method: $4x_1^{\text{new}} = 1 - x_2^{\text{old}} - 2x_3^{\text{old}}$
 $5x_2^{\text{new}} = 2 - 2x_1^{\text{old}} - x_3^{\text{old}}$
 $4x_3^{\text{new}} = 3 + x_1^{\text{old}} - 2x_2^{\text{old}}$

Gauss-Seidel Method: $4x_1^{\text{new}} = 1 - x_2^{\text{old}} - 2x_3^{\text{old}}$
 $5x_2^{\text{new}} = 2 - 2x_1^{\text{new}} - x_3^{\text{old}}$
 $4x_3^{\text{new}} = 3 + x_1^{\text{new}} - 2x_2^{\text{new}}$

Convergence Analysis.

If $\|S^{-1}T\|_{\infty} < 1$, then the iterative method will converge.

If A is strictly row diagonally dominant, then both iterative methods (Jacobi and Gauss-Seidel) converge.

Stopping Criterion. $\rightarrow \frac{\|x_k - x_{k-1}\|}{\|x_k\|} \leq \epsilon$ for some small $\epsilon > 0$.

Errors: $e_k = x_k - x$, $e_0 = x_0 - x$, then we have

$$\|e_k\|_{\infty} \leq \|S^{-1}T\|_{\infty} \|e_{k-1}\|_{\infty}$$

The Method of Successive Overrelaxation.

$$S = \begin{bmatrix} \sigma a_{11} & & \\ a_{21} & \sigma a_{22} & \\ \vdots & \vdots & \ddots \\ a_{n1} & a_{n2} & \dots & \sigma a_{nn} \end{bmatrix} \quad T = S - A$$

Let $A = L + D + U$, (L lower triangular, D diagonal, U strictly upper triangular Parts)

$$x^{new} = x^{old} + w D^{-1} [b - Lx^{new} - (D+U)x^{old}]$$

Where $w = \frac{1}{\sigma}$ is called the relaxation parameter. (When $w = 1$, we get the Gauss-Seidel Method)

The Advantage of Iterative Methods. (Page 11-12)

* Chapter 12. Determinants.

$$n=1: \det[a] = a.$$

$$n=2: \det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

$$n=3: \det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = a_{11} \det \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} - a_{12} \det \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix} + a_{13} \det \begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$n=4: \det \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} = a_{11} \det \begin{bmatrix} a_{22} & a_{23} & a_{24} \\ a_{32} & a_{33} & a_{34} \\ a_{42} & a_{43} & a_{44} \end{bmatrix} - a_{12} \det \begin{bmatrix} a_{21} & a_{23} & a_{24} \\ a_{31} & a_{33} & a_{34} \\ a_{41} & a_{43} & a_{44} \end{bmatrix} + a_{13} \det \begin{bmatrix} a_{21} & a_{22} & a_{24} \\ a_{31} & a_{32} & a_{34} \\ a_{41} & a_{42} & a_{44} \end{bmatrix} - a_{14} \det \begin{bmatrix} a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{bmatrix}$$

Property 1. $\det \begin{bmatrix} ca_{11} & ca_{12} & ca_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = c \det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ (or column)

2. $\det \begin{bmatrix} a'_{11} + a''_{11} & a'_{12} + a''_{12} & a'_{13} + a''_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \det \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} + \det \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ (or column)

3. If we interchange two rows (columns) of A , then the determinant will change by a minus sign.

4. If two of rows (columns) are identical, then $\det(A) = 0$

5. If we add/subtract a multiple of one row (column) of A to/from another row (column) of A , then the determinant is unchanged.

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \quad B = \begin{bmatrix} a & b & c \\ d+ka & e+kb & f+kc \\ g & h & i \end{bmatrix} \Rightarrow \det A = \det B.$$

6. $\det A = \det A^T$

7. If A is a triangular matrix, then $\det A$ is just the product of the diagonal components.

8. (Multiplicative Property)

$$\det(AB) = \det A \det B \quad \text{where } A, B \text{ } n \times n \text{ matrices.}$$

9. (Cofactor Expansion)

$$\begin{aligned} \det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} &= a \det \begin{bmatrix} e & f \\ h & i \end{bmatrix} - b \det \begin{bmatrix} d & f \\ g & i \end{bmatrix} + c \det \begin{bmatrix} d & e \\ g & h \end{bmatrix} \rightarrow \text{1st row} \\ &= -d \det \begin{bmatrix} b & c \\ h & i \end{bmatrix} + e \det \begin{bmatrix} a & c \\ g & i \end{bmatrix} - f \det \begin{bmatrix} a & b \\ g & h \end{bmatrix} \rightarrow \text{2nd row} \\ &= a \det \begin{bmatrix} e & f \\ h & i \end{bmatrix} - d \det \begin{bmatrix} b & c \\ h & i \end{bmatrix} + g \det \begin{bmatrix} b & c \\ e & f \end{bmatrix} \rightarrow \text{1st column} \\ &= -b \det \begin{bmatrix} d & f \\ g & i \end{bmatrix} + e \det \begin{bmatrix} a & c \\ g & i \end{bmatrix} - h \det \begin{bmatrix} a & c \\ d & f \end{bmatrix} \rightarrow \text{2nd column} \end{aligned}$$

Signs

$$n=4 \quad \begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix}$$

$$n=3 \quad \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$